

| Question |     |      |  | Marking details                                                                                                                                                                                                                 | Marks available |            |            |       |            |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|------------|-------|------------|------|
|          |     |      |  |                                                                                                                                                                                                                                 | AO1             | AO2        | AO3        | Total | Maths      | Prac |
| 11.      | (a) |      |  | moles water = $1.44 \div 18.02 = 0.0799$ (1)<br><br>d = 4 (1) [allow ecf]                                                                                                                                                       |                 | 1          | 1          | 2     | 2          | 2    |
|          | (b) | (i)  |  | bromide is present (1)<br><br>the sample produces an orange-brown solution which could be bromine or iodine but cannot contain iodide as no H <sub>2</sub> S produced (1)                                                       |                 |            | 1<br><br>1 | 2     |            | 2    |
|          |     | (ii) |  | moles Br = 0.0203 (1) [allow moles I = 0.01623 as ecf]<br><br>moles compound = $5.076 \times 10^{-3}$ so c = 4 (1)<br>[allow 3.2 as ecf for iodide]                                                                             |                 | 1          | 1          | 2     | 2          | 2    |
|          | (c) |      |  | aluminium is amphoteric                                                                                                                                                                                                         | 1               |            |            | 1     |            | 1    |
|          | (d) |      |  | $6.34\% \times 425.62 = 27.0$ shows b = 1 (1)<br><br>$1.63\% \times 425.62 = 6.94$ so must be Li and only one present (1)<br><br>therefore formula is LiAlBr <sub>4</sub> .4H <sub>2</sub> O [allow ecf from earlier parts] (1) |                 | 1<br><br>1 | 1          | 3     | 1<br><br>1 |      |
|          |     |      |  | Question 11 total                                                                                                                                                                                                               | 1               | 4          | 5          | 10    | 6          | 7    |